Logistics, Delivery & Returns Insights

How well are we delivering, and what causes returns?

Capstone Project, Option B | Python Programming Data Analyst Professional Certificate

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1. Project Overview

Business context

An online retailer sells products in 8 categories and delivers orders through 5 couriers (2GO, LBC, GrabExpress, J&T and NinjaVan). Management wants to know whether deliveries are working and why customers send products back.

Objectives

Answer one question: how well are we delivering, and what causes returns? To answer it, the project:

- measures delivery performance (lead time and shipment status),
- compares the five couriers,
- measures the return rate and the reasons for returns, by courier and by category,
- turns the findings into practical recommendations.

Scope

- **Included:** 7 data tables (customers, products, orders, order items, payments, shipments, returns), 15,000 orders from January 2024 to January 2026, 15,000 shipments and 1,200 returns.
- **Not included:** revenue and profit analysis, customer segments, the time between order and shipping (the dates are not reliable), refund amounts, and statistical tests.

Approach

Python (pandas, numpy, matplotlib and seaborn) in a Jupyter notebook. The data was cleaned using new _clean columns, so the original data stays untouched, and every fix is recorded in a 35-line cleaning log.

Executive Summary

- This project looks at how well we deliver orders and what causes returns, using 15,000 orders from 7 tables.
- I cleaned the data first (documented in a 35-line cleaning log, original files untouched), then built the analysis in Python.
- 77.87% of shipments were delivered, 17.96% are still in transit, and 4.17% were returned to sender or lost.
- Delivered shipments take 1 to 7 days, about 4.00 days on average, and all five couriers are within 0.06 days of each other.
- 1,200 of 15,000 orders were returned, an overall return rate of 8.00%.
- NinjaVan (8.81%), J&T (8.61%) and GrabExpress (8.44%) are above the overall rate, while 2GO (7.01%) and LBC (7.11%) are below it.
- J&T also has the lowest delivered rate (76.59%) and the highest returned to sender rate (3.58%).
- 70.33% of returns (844) come from reasons we control: defective, missing parts, late delivery, wrong item and damaged item.

- Return rates are similar across categories (7.47% to 8.65%), so category is not the main driver.
- Top recommendation 1: add a quality and packing check before dispatch to reduce the returns we cause.
- Top recommendation 2: track return rate and lost shipments for each courier every month, and review NinjaVan, J&T and GrabExpress.
- Three more recommendations: reduce lost deliveries (starting with J&T), set a delivery time target, and improve date and refund data.
- Data note: the data is synthetic and some dates are out of order (for example, shipped before ordered), so lead time results need care.

Notebook screenshots

Screenshots of the notebook (code and output) show how the analysis was built. Each screenshot below is taken from the final notebook, Capstone_OptionB_Logistics_Returns.ipynb.

Notebook screenshot 1: delivery KPIs (Section 4.1, code and output)

4. Analysis

4.1 Delivery KPIs

In [99]:
```
# how many shipments have each status, value_counts() counts them
status_counts = shipments["shipment_status"].value_counts()
```

In [100…
```
# percent = count divided by all shipments times 100. round(2) keep 2 decimals
status_percent = (status_counts / len(shipments) * 100).round(2)
```

In [101…
```
# put the 2 together in one small table
status_table = pd.DataFrame({"shipments": status_counts, "percent": status_percent})
display(status_table)
print("total shipments:", len(shipments))
```

shipment_status	shipments	percent
Delivered	11680	77.87
In Transit	2694	17.96
Returned to Sender	465	3.10
Lost	161	1.07

```
total shipments: 15000
```

In [102…
```
# lead time
# lead time only exists for delivered shipments, so these numbers use only those
# mean() is the average, median() is the middle value
print("shipments with a lead time:", shipments["lead_time_days"].notna().sum())
print("average lead time:", shipments["lead_time_days"].mean().round(2), "days")
print("median lead time:", shipments["lead_time_days"].median(), "days")
print("fastest:", shipments["lead_time_days"].min(), "days, slowest:", shipments["lead_time_days"].max(), "days")
```

```
shipments with a lead time: 11122
average lead time: 4.0 days
median lead time: 4.0 days
fastest: 1.0 days, slowest: 7.0 days
```

Notebook screenshot 2: courier comparison (Section 4.2, code and output)

4.2 Courier comparison

In [103…

```
#Courier
# how many shipments each courier has?
courier_count = shipments["courier"].value_counts()

# groupby() makes one group for each courier, then I take the average lead time of each group
courier_lead = shipments.groupby("courier")["lead_time_days"].mean().round(2)

# put them together, sort_values() puts the fastest courier first
courier_table = pd.DataFrame({"shipments": courier_count, "avg_lead_time_days": courier_lead})
courier_table = courier_table.sort_values("avg_lead_time_days")
display(courier_table)
```

courier	shipments	avg_lead_time_days
NinjaVan	2985	3.97
LBC	3011	3.98
2GO	2951	4.01
J&T	3042	4.02
GrabExpress	3011	4.03

In [104…

```
# what happens to each courier's shipments
#crosstab() counts the statuses for each courier
# normalize="index" turns each courier row into shares that add up to 1, so I times 100 to get percent
status_by_courier = pd.crosstab(shipments["courier"], shipments["shipment_status"], normalize="index") * 100
status_by_courier = status_by_courier.round(2)
display(status_by_courier)
```

courier	shipment_status	Delivered	In Transit	Lost	Returned to Sender
2GO		77.43	18.43	1.05	3.08
GrabExpress		78.31	17.60	1.16	2.92
J&T		76.59	18.57	1.25	3.58
LBC		78.55	17.60	1.06	2.79
NinjaVan		78.46	17.59	0.84	3.12

Notebook screenshot 3: Chart 3, return rate by courier (code above, chart below)

In [118…

```
# chart 3: return rate by courier
fig, ax = plt.subplots(figsize=(8, 5.5))

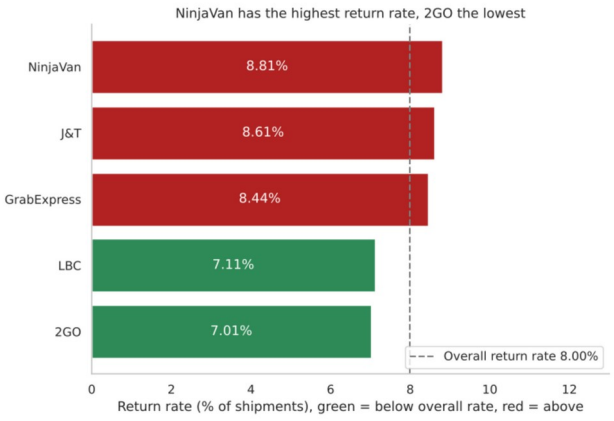
# np.where checks every bar: red if its rate is above the overall rate, green if not
bar_colors = np.where(courier_returns["return_rate_pct"] > return_rate, red, green)
ax.barh(courier_returns.index, courier_returns["return_rate_pct"], color=bar_colors)

# write the rate inside each bar, in white
labels = [f"{r:.2f}%" for r in courier_returns["return_rate_pct"]]
ax.bar_label(ax.containers[0], labels=labels, label_type="center", color="white")

# dashed line = overall return rate of all orders
ax.axvline(return_rate, color="gray", linestyle="--", label="Overall return rate 8.00%")
ax.legend(loc="lower right")

ax.set_xlim(0, 13)
ax.set_title("NinjaVan has the highest return rate, 2GO the lowest")
ax.set_xlabel("Return rate (% of shipments), green = below overall rate, red = above")
ax.grid(False)
ax.spines[["top", "right"]].set_visible(False)

plt.tight_layout()
plt.savefig(CHARTS / "chart3_return_rate_courier.png", dpi=150)
plt.show()
```



Notebook screenshot 4: cleaning log (Section 3.11, 35 lines)

	table	what I fixed	rows before	rows after
0	customers	Removed duplicate rows	5050	5000
1	orders	Removed duplicate rows	15075	15000
2	order_items	Removed duplicate rows	37763	37501
3	products	Checked duplicates, none found	800	800
4	shipments	Checked duplicates, none found	15000	15000
5	returns	Checked duplicates, none found	1200	1200
6	payments	Checked duplicates, none found	15000	15000
7	orders	Parsed mixed dates into order_datetime_clean	15000	15000
8	shipments	Parsed mixed dates into ship_date_clean and delivered_date_clean	15000	15000
9	returns	Parsed mixed dates into return_request_date_clean	1200	1200
10	products	Standardized 18 category spellings into 8 (category_clean)	800	800
11	products	Removed peso signs and commas, made unit_price and unit_cost numbers (_clean columns)	800	800
12	shipments	Lead time only for Delivered shipments, 3149 other shipments had delivered dates that cannot be trusted	15000	15000
13	orders	Flagged 43 orders with unknown customer CUST999999 (customer_orphan), kept for delivery and returns analysis	15000	15000
14	order_items	Flagged 129 items with unknown product PROD00000 (product_orphan), left out of category analysis only	37501	37501
15	order_items	Removed 372 order lines with quantity 0 and made quantity_clean (2 of them were product orphans, so 127 orphan items remain)	37501	37129
16	order_items	Removed peso signs and commas in unit_price_at_purchase (unit_price_at_purchase_clean)	37129	37129
17	order_items	Filled 2217 blank prices with the catalog price of the same product (price_was_blank marks them), 9 stay blank because the product is unknown. Non blank prices matched the catalog 100 percent	37129	37129
18	orders	23 orders have no order lines left after the quantity 0 removal, kept for delivery and returns analysis	15000	15000
19	orders	Made discount_pct and shipping_fee numbers (_clean columns), 756 and 470 blanks kept as blank, not guessed	15000	15000
20	returns	Made refund_amount a number (refund_amount_clean), 187 blanks kept as blank	1200	1200
21	returns	Filled 110 blank refunded_flag with Unknown (refunded_flag_clean)	1200	1200
22	products	Made stock_qty a number (stock_qty_clean, 19 blanks kept) and merged pc, pcs, piece into pc (uom_clean)	800	800
23	customers	Standardized gender (M, F, Male, Female, female, blank) into Male, Female, Unknown (gender_clean)	5000	5000
24	customers	Made age a number (age_clean), 203 ages under 13 (48 of them negative) set to blank, 149 were already blank	5000	5000
25	customers	Filled 264 blank loyalty_tier with Unknown (loyalty_tier_clean)	5000	5000
26	customers	Checked emails, 225 wrong ones (missing @ or dot) and 104 blanks, email_clean keeps only good emails and email_valid shows which ones are good	5000	5000
27	customers	Checked phones, 133 short numbers made blank in phone_clean (phone_valid marks the good ones, 11 digits starting with 09), 145 were already blank	5000	5000
28	customers	Parsed 3 mixed signup_date styles into signup_date_clean (slash dates read as month first)	5000	5000
29	payments	Removed peso signs and commas in amount_paid and made it a number (amount_paid_clean), 878 blanks kept as blank, 657 amounts are 0 and kept as they are	15000	15000
30	payments	Parsed mixed payment_date into payment_date_clean (slash dates read as month first)	15000	15000
31	shipments	Checked order dates: 7397 shipments have a ship_date before the order date (21 on the same day), kept as they are, lead time uses ship date to delivered date so it is not affected, order to ship time is not analyzed	15000	15000
32	returns	Checked return dates: 583 returns were requested before the order date, kept as they are and noted as a data limitation	1200	1200
33	returns	Checked refunds: 248 returns with refunded_flag No have a refund amount and 99 with Yes have none, kept as they are, refund amounts are not used for the return rate	1200	1200
34	returns	Return rate is counted from the returns table (1200 returns, one per order) because order_status Returned only matches 40 of them	1200	1200

```
lines in my log: 35
```

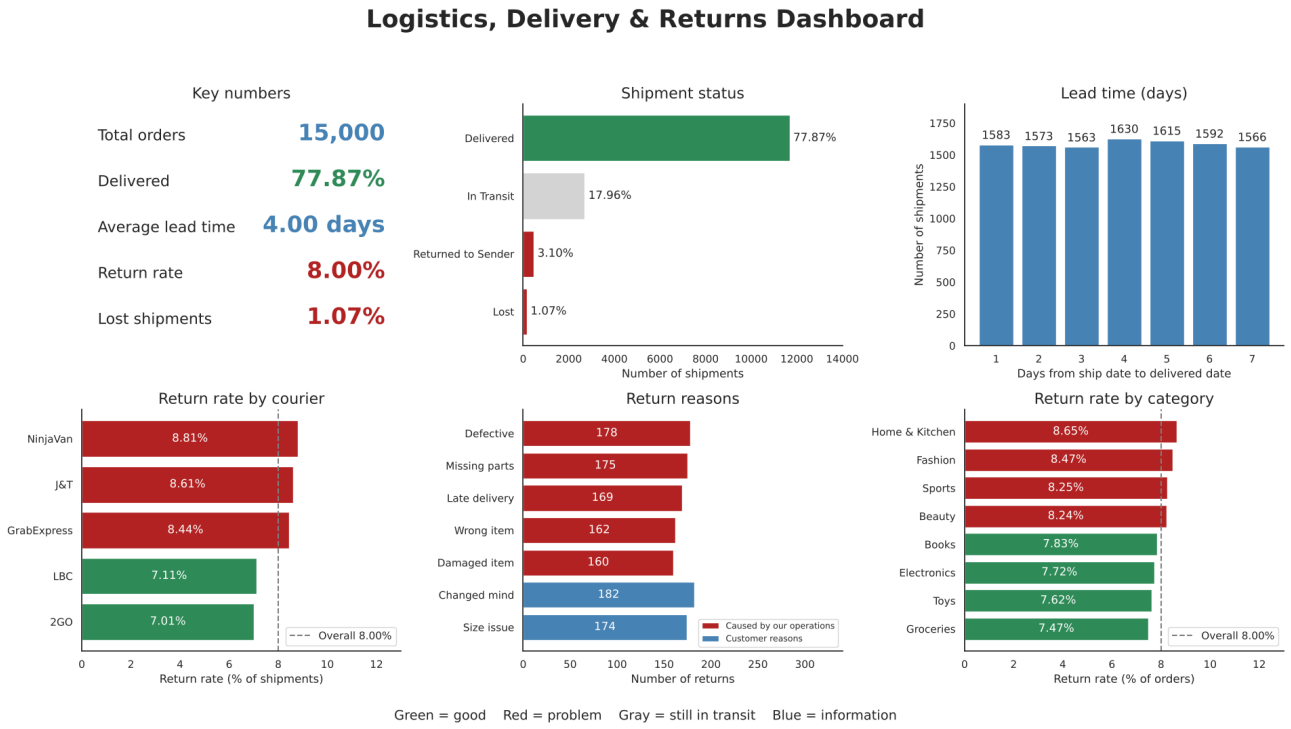
Cleaned data files

The cleaned tables are saved in the data_clean folder. The original columns are kept, and the cleaned versions are added as _clean columns next to them.

File	Rows	Columns	What was cleaned
cleaned_customers.csv	5,000	21	Duplicates, gender, age, loyalty tier, email, phone, signup date
cleaned_products.csv	800	14	Category spellings, prices, unit cost, stock quantity, unit of measure
cleaned_orders.csv	15,000	12	Duplicates, dates, discount, shipping fee, orphan flag
cleaned_order_items.csv	37,129	9	Duplicates, quantity 0, prices, orphan flag
cleaned_payments.csv	15,000	8	Amounts and payment dates
cleaned_shipments.csv	15,000	10	Dates and lead time
cleaned_returns.csv	1,200	9	Dates, refund amount, refund flag
cleaning_log.csv	35	4	Every fix, with rows before and after

2. Dashboard Screenshot

The dashboard brings the key numbers and the main charts onto one page. It is saved at 1920 x 1080 pixels (16:9).



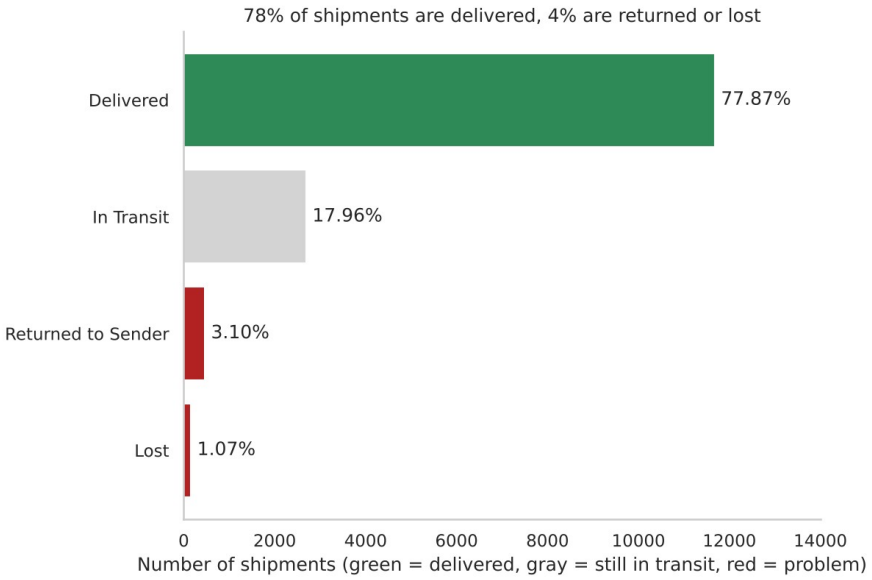
What it shows: The key numbers and the five main charts on one page: 77.87% delivered, 4.00 days average lead time, 8.00% return rate and 1.07% lost.

So what: Delivery is stable and the main chance to improve is returns, especially the 70.33% that we cause and the three couriers above the average return rate.

Exported charts

Color rules used in every chart: **green** = good, **red** = problem, **gray** = still in transit, **blue** = information.

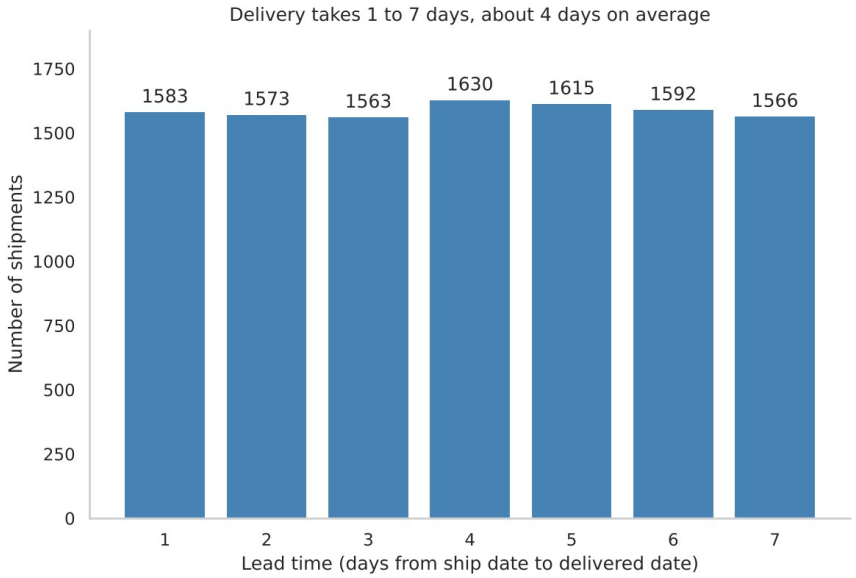
Chart 1: Shipment status



What it shows: 77.87% of shipments (11,680 of 15,000) were delivered and 17.96% are still in transit. Only 4.17% (626 shipments) were returned to sender or lost.

So what: Most deliveries work. The red bars are small, but each lost or returned shipment is a failed delivery that costs us money and a customer.

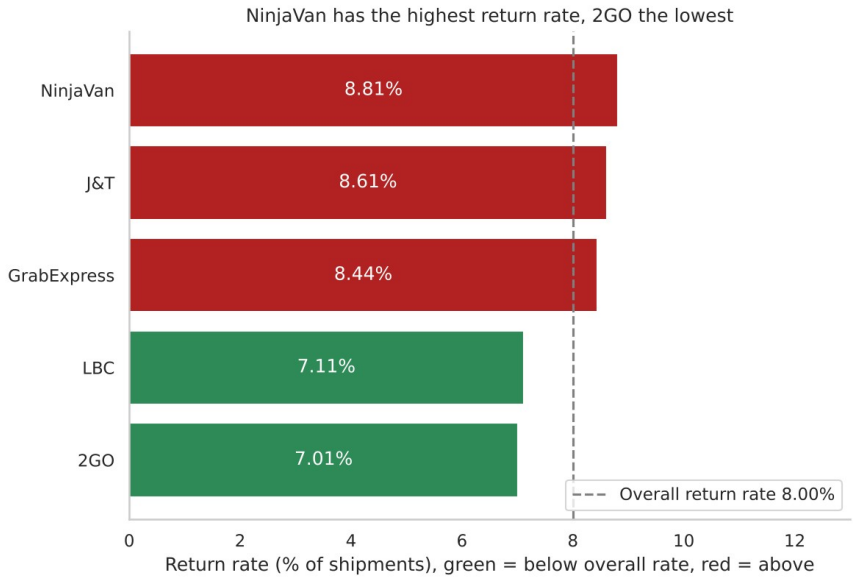
Chart 2: Lead time



What it shows: Delivery takes 1 to 7 days, about 4.00 days on average. Every day count has a similar number of shipments (1,563 to 1,630). This uses the 11,122 shipments that have a lead time.

So what: Delivery time is predictable, but there is no "fast" group. A 7-day delivery is as common as a 1-day one.

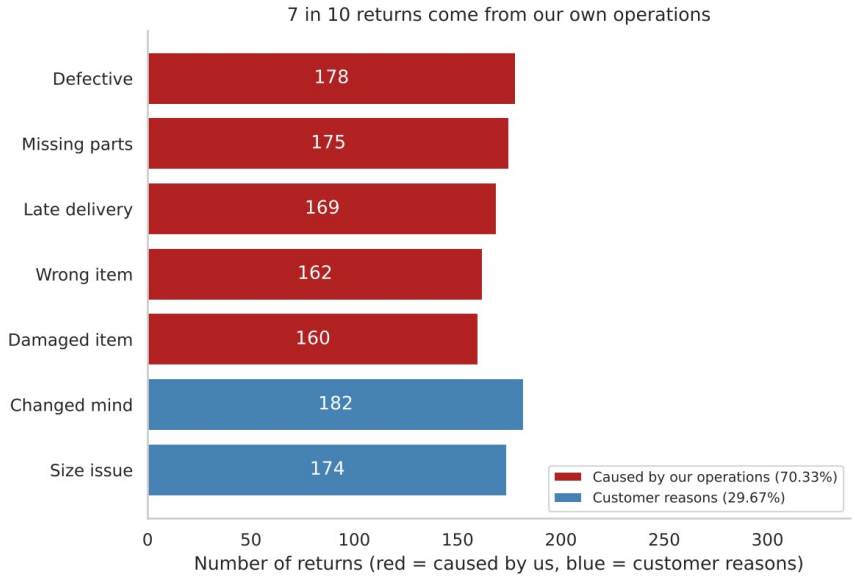
Chart 3: Return rate by courier



What it shows: NinjaVan (8.81%), J&T (8.61%) and GrabExpress (8.44%) are above the overall 8.00% return rate. 2GO (7.01%) and LBC (7.11%) are below it.

So what: The couriers deliver at the same speed (3.97 to 4.03 days), so the return rate is what separates them. The gap between the highest and lowest is 1.80 points.

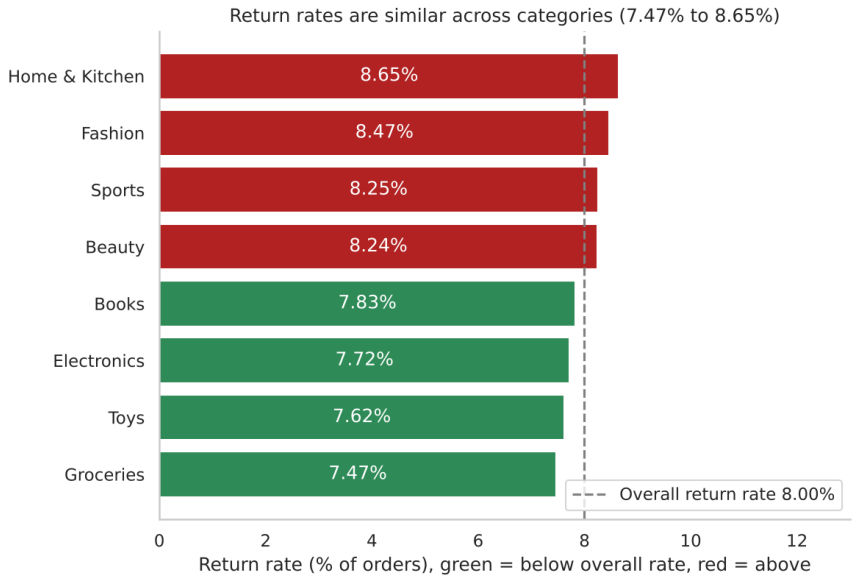
Chart 4: Return reasons



What it shows: 844 of 1,200 returns (70.33%) come from reasons we control: defective, missing parts, late delivery, wrong item and damaged item. The two customer reasons (changed mind, size issue) make up 29.67%.

So what: No single reason dominates (each is between 160 and 182 returns). That points to a general quality and packing problem, not one specific fix.

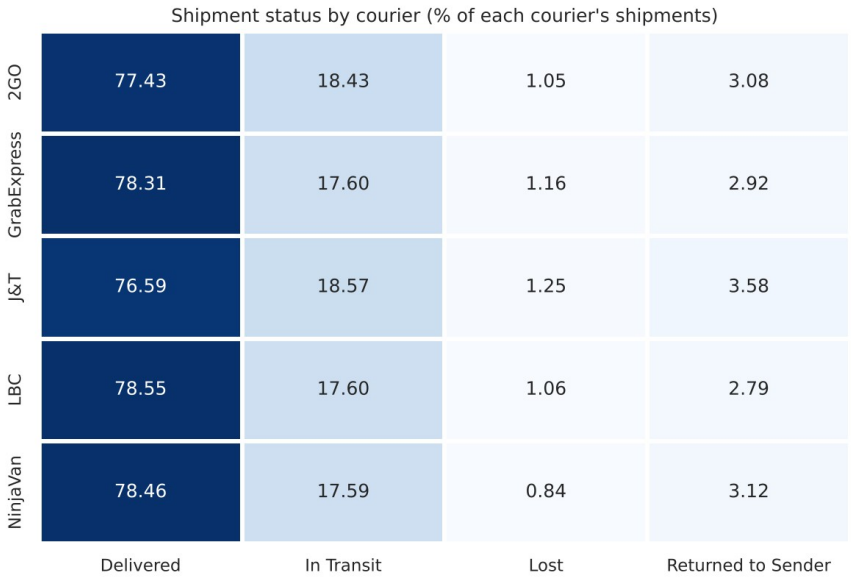
Chart 5: Return rate by category



What it shows: Return rates range from 7.47% (Groceries) to 8.65% (Home & Kitchen). Four categories are above the 8.00% overall rate and four are below it.

So what: The gap between categories is small (1.18 points), so category is not the main driver of returns.

Chart 6: Shipment status by courier (heatmap)



What it shows: Delivered is between 76.59% and 78.55% for every courier. J&T has the lowest delivered rate (76.59%), the highest lost rate (1.25%) and the highest returned to sender rate (3.58%).

So what: Couriers are close on most numbers, but J&T is the weakest on every delivery problem, so it is the first courier to review.

3. Data Analysis & Methodology

Data and tools

7 CSV files (customers, products, orders, order items, payments, shipments, returns) covering 15,000 orders from January 2024 to January 2026. Python (pandas, numpy, matplotlib, seaborn) in a Jupyter notebook.

Cleaning approach

I never changed the original files or columns. Every fix went into a new _clean column, and 35 cleaning steps are recorded in a cleaning log. The main steps:

- Removed duplicate rows (50 customers, 75 orders, 262 order lines). A duplicate means a row where every column is identical to another row, and I kept the first one.
- Standardized mixed date formats (slash dates are month first).
- Standardized 18 product category spellings into 8.
- Removed peso signs and commas from prices and turned them into numbers.
- Removed 372 order lines with quantity 0.
- Filled 2,217 blank order prices with the catalog price. The 34,794 prices that were not blank matched the catalog 100%.
- Flagged orphan records (43 orders with an unknown customer, 129 order lines with an unknown product) instead of deleting them.
- Left other blanks as blank, because I did not want to guess values.

Blank values: fill or leave blank?

I only filled a blank when the true value could be recovered from another reliable source. For example, 2,217 blank order prices were filled with the catalog price of the same product (the 34,794 prices that were not blank matched the catalog 100%), and blank gender, loyalty tier and refunded flag were labeled Unknown so those rows stay in the counts.

I left amounts and dates blank (discount, shipping fee, amount paid, refund amount, delivered date, and invalid ages, emails and phone numbers), because there is no source for the true value. None of these columns is needed for the delivery and return results, and each decision is recorded in the cleaning log.

A blank is honest, and a guessed number looks real but isn't.

Key metrics and calculations

Metric	How it is calculated
Delivered rate	Delivered shipments divided by all shipments, times 100
Lead time	Delivered date minus ship date, in days
Average lead time by courier	Average lead time for each courier
Return rate	Returns divided by orders, times 100 (1,200 of 15,000)
Return rate by courier	Returns per courier divided by shipments per courier
Return rate by category	Returned orders containing the category divided by orders containing the category
Share of returns by reason	Returns per reason divided by all returns

Filters and rules

- Lead time is only calculated for shipments with status Delivered (11,122 shipments).
- Order lines with an unknown product are left out of the category analysis only (127 lines).
- An order is counted once per category, even if it has two lines of the same category.
- Returns caused by our operations are: defective, missing parts, late delivery, wrong item and damaged item. Changed mind and size issue are customer reasons.

Analytical techniques

Counting and percentages (value_counts), grouping (groupby), cross tables (crosstab), lookups between tables (map, like VLOOKUP), joining tables into one fact table with one row per order (merge, 15,000 rows and 18 columns), and comparing each courier and category with the overall rate. NumPy (np.where) picks the bar color in the charts: red when a rate is above the overall rate, green when it is below.

Limitations

The data is synthetic. 558 delivered shipments have no delivered date. Many shipments (7,397) are dated before their order date, so the dates are not fully reliable. This is a descriptive analysis: it shows where returns happen, but it does not prove the cause.

4. Key Takeaways

- **Delivery works.** 77.87% of shipments are delivered, in about 4 days on average.
- **Returns are the bigger issue.** 8.00% of orders are returned, and 70.33% of those returns come from reasons we control.
- **Couriers differ on returns, not on speed.** NinjaVan has the highest return rate (8.81%) and 2GO the lowest (7.01%).
- **J&T is the weakest courier on delivery.** It has the lowest delivered rate (76.59%) and the highest lost rate (1.25%).
- **Product category is not the main driver of returns.** Rates range from 7.47% to 8.65%.

5. Insights

1. **Most shipments are delivered.** 11,680 of 15,000 shipments (77.87%) were delivered, 17.96% are still in transit, and 4.17% (626 shipments) were returned to sender or lost.
2. **Delivery time is steady.** Delivered shipments take 1 to 7 days, about 4.00 days on average, and each day count has a similar number of shipments (1,563 to 1,630).
3. **Couriers are equal on speed.** Average lead time is between 3.97 and 4.03 days for every courier, so speed does not separate them. Returns do.
4. **8.00% of orders are returned.** 1,200 of 15,000 orders came back.
5. **Three couriers are above the average return rate.** NinjaVan (8.81%), J&T (8.61%) and GrabExpress (8.44%) are above 8.00%. 2GO (7.01%) and LBC (7.11%) are below it. The gap between the highest and lowest is 1.80 points.
6. **J&T has the weakest delivery results.** It has the lowest delivered rate (76.59%) and a lost rate of 1.25%, compared with 0.84% for NinjaVan, the lowest.
7. **Most returns are caused by us.** 844 of 1,200 returns (70.33%) are for reasons tied to our own operations: defective, missing parts, late delivery, wrong item and damaged item. Only 356 (29.67%) are customer reasons (changed mind, size issue).
8. **Category is not the main driver.** Return rates are close across categories, from 7.47% (Groceries) to 8.65% (Home & Kitchen).

6. Summary

This project looked at how well the company delivers orders and what causes returns, using 15,000 orders across 7 tables. After cleaning the data, I measured delivery performance, compared 5 couriers and analyzed 1,200 returns by reason, courier and category. Delivery is steady: 77.87% of shipments are delivered in about 4 days, and only 4.17% are returned to sender or lost. The main problem is returns. 8.00% of orders come

back, and 70.33% of those are caused by our own operations, such as defective items, missing parts and wrong items. Return rates differ by courier but are similar across categories.

7. Conclusions

- **How well are we delivering?** Well overall. Delivery time is stable, and 77.87% of shipments are delivered. The weak spot is J&T and the 4.17% of shipments that are lost or returned to sender.
- **What causes returns?** Mostly our own operations (70.33%), not the product category or delivery speed. The courier also matters: the gap between the highest and lowest return rate is 1.80 points.
- **What to do first:** add a quality and packing check before dispatch, and review the three couriers with above-average return rates, starting with J&T.
- **What to remember:** the data is synthetic and has date problems, so these are patterns to investigate, not proven causes.

8. Recommendations

1. **Reduce the returns we cause (top priority)**
   - **Based on:** 844 of 1,200 returns (70.33%) come from defective items, missing parts, late delivery, wrong item and damaged item. No single reason dominates (160 to 182 returns each), which points to a general quality and packing problem.
   - **Action:** Add a quality and packing check before dispatch: item condition, correct item, complete parts, protective packing. Check the products and suppliers that appear most often in defective and damaged returns.
   - **How to track:** Count operations-caused returns every month. For example, cutting them by one third would bring the overall return rate from 8.00% to about 6.1%.
2. **Review the couriers with above-average return rates (top priority)**
   - **Based on:** NinjaVan (8.81%), J&T (8.61%) and GrabExpress (8.44%) are above the 8.00% average. 2GO (7.01%) and LBC (7.11%) are below it. All couriers deliver at the same speed (3.97 to 4.03 days), so returns are what separates them.
   - **Action:** Build a monthly courier scorecard with return rate, delivered rate, lost rate and lead time. Meet with the three couriers above average. If 2GO and LBC keep lower rates, move more orders to them step by step.
   - **How to track:** Compare each courier's return rate with the 8.00% overall rate every month.
3. **Reduce lost and failed deliveries, starting with J&T**
   - **Based on:** 626 shipments (4.17%) were lost or returned to sender. J&T has the lowest delivered rate (76.59%), the highest lost rate (1.25%) and the highest returned to sender rate (3.58%).
   - **Action:** Raise the J&T results directly with J&T. Set up a claim process for lost parcels and check delivery addresses before dispatch.
   - **How to track:** Watch the lost and returned to sender percentages for each courier every month.
4. **Set a delivery time target**
   - **Based on:** Delivery takes 1 to 7 days, and about 43% of delivered shipments (4,773 of 11,122) take 5 to 7 days. Late delivery is also the reason for 169 returns (14.08%).
   - **Action:** Set a target (for example, 5 days or less) and flag shipments that go past it, so they can be followed up before the customer returns the order.
   - **How to track:** The share of shipments delivered within the target, and the number of late delivery returns.

5. **Improve data quality so the next analysis is more reliable**
    - **Based on:** 558 delivered shipments have no delivered date, 7,397 shipments are dated before their order date, 583 returns are dated before their order date, and 347 refund records do not match the refund flag (248 and 99).
    - **Action:** Make delivered date and refund amount required fields, and check that dates are in a logical order when records are saved.
    - **How to track:** The number of records that fail these checks each month, which should go down.

What not to prioritize: product category. The return rate only ranges from 7.47% to 8.65% (a 1.18 point gap), so fixing one category will not solve the returns.